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1. Introduction

Artificial Intelligence and natural language processing technologies have significantly improved the way users interact with digital information systems. Among recent advancements, Retrieval-Augmented Generation (RAG) has emerged as an effective approach for combining information retrieval techniques with large language models to generate context-aware and document-grounded responses.

This project presents **EduRAG**, an intelligent educational assistant designed to support educational question answering through document-based response generation. The system integrates document retrieval and language generation within an interactive prototype that allows users to upload educational materials and receive responses based on retrieved document content.

2. Description of the Identified Issue

The increasing availability of digital educational resources has created challenges in efficiently accessing accurate information from large educational documents. Students often need to manually search through lengthy textbooks and technical materials to find specific concepts or explanations, which can be time-consuming and inefficient. In addition, general conversational AI systems may generate responses that are not grounded in trusted educational sources, leading to inaccurate or unsupported answers. Ensuring that generated responses are both relevant and supported by reliable educational content has therefore become an important challenge in educational question-answering systems.

To address this issue, this project focuses on developing a Retrieval-Augmented Generation (RAG) system capable of retrieving relevant information from educational documents and generating context-aware responses through an interactive and user-friendly prototype.

3. Proposed Solution and System Architecture

3.1 Proposed Solution

The proposed solution is **EduRAG**, an educational Retrieval-Augmented Generation system designed to improve document-based question answering. The system combines document retrieval techniques with large language model-based response generation to produce context-aware responses grounded in educational resources. The system enables users to upload educational PDF documents, submit natural language queries, retrieve relevant document content, and receive generated responses through an interactive Streamlit-based interface.

3.2 System Architecture

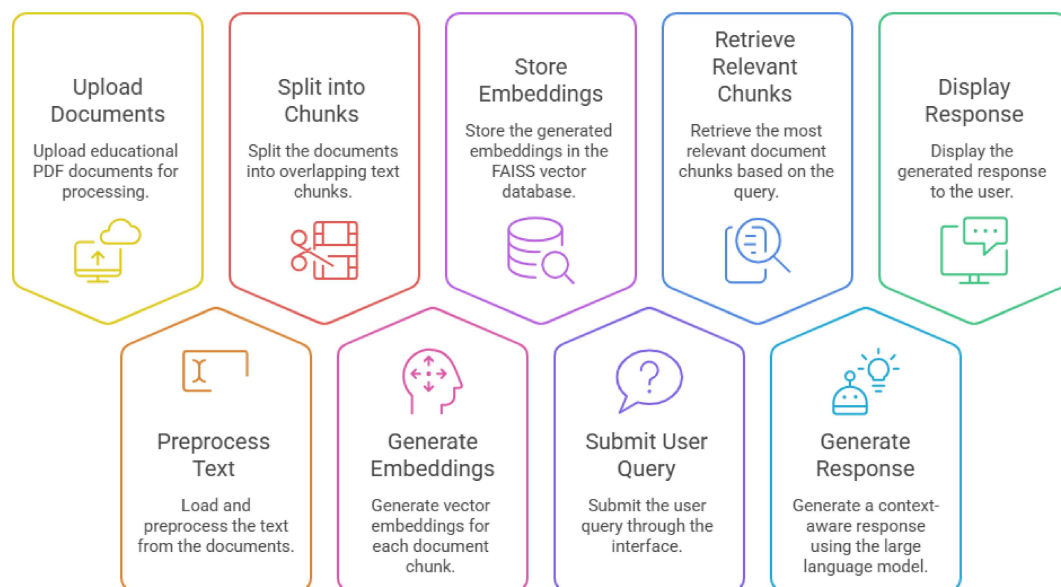
The architecture of the proposed system follows an end-to-end Retrieval-Augmented Generation pipeline. First, educational PDF documents are loaded and preprocessed before being divided into overlapping text chunks. The processed chunks are then

converted into vector representations using a custom lightweight embedding mechanism.

The generated embeddings are stored within a FAISS vector database to support similarity-based retrieval. When a user submits a query, the system retrieves the most relevant document chunks and forwards them to the Groq large language model to generate a context-aware response. Finally, the generated response is displayed through the interactive user interface.

3.3 System Workflow

The overall workflow of the proposed **EduRAG** system is illustrated in Figure 1.



4. Description of Models or RAG Components

4.1 Document Processing and Chunking

The **EduRAG** system begins with a document processing stage responsible for extracting and preparing educational content for retrieval operations. Educational PDF documents are first loaded using a document loader module that extracts textual content from the uploaded files.

After document loading, the extracted text undergoes preprocessing before being divided into smaller overlapping text chunks. Document chunking is applied to improve retrieval efficiency and preserve contextual continuity between text segments. The system uses overlapping chunks to reduce information loss between consecutive text sections and improve the quality of retrieved content during similarity-based search operations.

4.2 Embedding and Vector Storage

After document chunking, the generated text chunks are transformed into vector representations using a custom lightweight embedding mechanism. The embedding process converts textual content into vector representations that enable similarity-based comparison between user queries and document chunks.

The proposed system applies a character-frequency-based embedding approach to generate normalized vector representations while maintaining a lightweight and efficient retrieval pipeline. After embedding generation, all document vectors are stored within a FAISS vector database to support efficient similarity search and retrieval operations across the processed educational documents.

4.3 Retrieval Mechanism

The retrieval mechanism is responsible for identifying the most relevant document content related to the user query. When a query is submitted through the system interface, it is converted into a vector representation using the same embedding mechanism applied during document processing.

The generated query vector is compared with the stored document embeddings using similarity-based search operations within the FAISS vector database. The system then retrieves the most relevant document chunks, which are forwarded to the large language model as contextual input for response generation. This retrieval process helps improve response relevance by grounding generated answers in educational document content.

4.4 Large Language Model

The **EduRAG** system uses the **Groq** language model service to generate context-aware responses based on retrieved educational content. After the retrieval stage, the retrieved document chunks are provided to the language model as contextual input for response generation.

The system utilizes the *llama-3.3-70b-versatile* model to generate natural language responses grounded in the retrieved educational material. By combining retrieved document context with language model generation, the system improves response relevance and supports educational question answering.

4.5 User Interface

The **EduRAG** system includes an interactive user interface developed using the Streamlit framework. The interface enables users to upload educational PDF documents, submit natural language queries, and view generated responses in real time.

The interface also supports document interaction and response visualization within a unified environment, providing a simple and user-friendly educational question-answering experience.

5. Dataset Description

The proposed **EduRAG** system utilizes a collection of educational PDF documents related to deep learning and artificial intelligence concepts. These documents serve as the primary knowledge source for retrieval and response generation within the Retrieval-Augmented Generation pipeline.

The document collection includes educational resources such as *Dive into Deep Learning* and *The Little Book of Deep Learning*. The selected documents contain educational content covering machine learning fundamentals, neural network architectures, optimization techniques, natural language processing, and deep learning applications. These resources were selected because they provide comprehensive educational material suitable for document-based question answering and educational information retrieval.

The documents are processed through the document loading and preprocessing pipeline before being divided into overlapping text chunks for retrieval operations. After preprocessing and chunking, the generated text chunks are transformed into vector representations and stored within the FAISS vector database to support similarity-based retrieval and context-aware response generation.

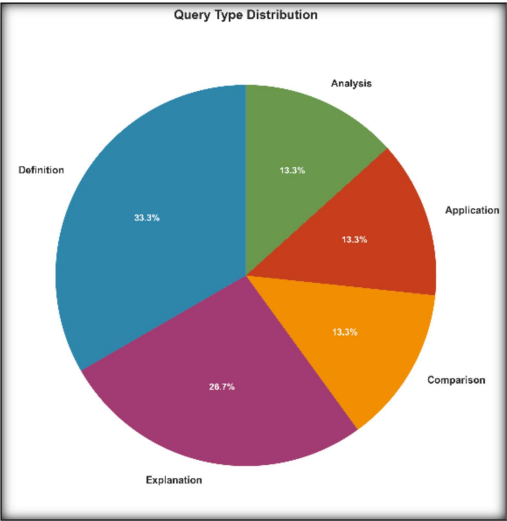
6. Evaluation Methodology and Result

6.1 Evaluation Methodology

The **EduRAG** system was evaluated using multiple educational queries designed to assess retrieval performance, response quality, and system efficiency. The evaluation process included different query categories such as definition, explanation, comparison, application, and analysis questions to examine the system's ability to handle diverse educational requests.

During the evaluation process, system performance was measured using response time metrics, retrieval quality measurements, and response generation analysis. A total of 15 educational queries were tested to evaluate the effectiveness of the Retrieval-Augmented Generation pipeline.

Figure 2. Query Type Distribution



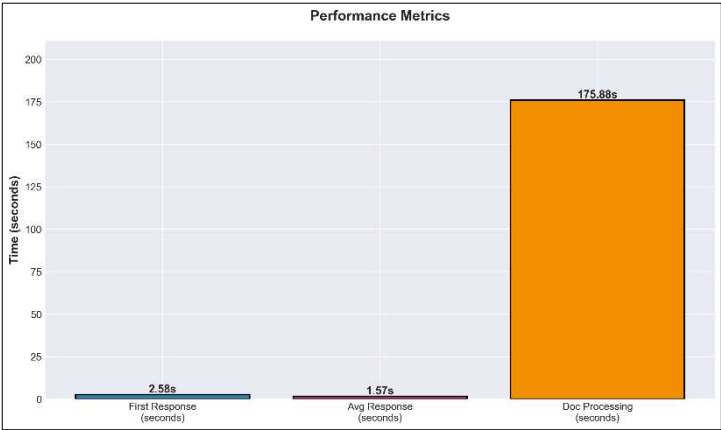
The figure illustrates the evaluated educational query categories..

6.2 Performance Results

The performance evaluation results demonstrate that the **EduRAG** system achieved stable response generation and efficient retrieval performance during testing. The evaluation focused on response generation speed, average query processing time, and document processing efficiency.

The system achieved a first response time of 2.58 seconds and an average response time of 1.57 seconds during query evaluation. In addition, the document processing pipeline required approximately 175.88 seconds for document loading, preprocessing, chunking, and vector database generation.

Figure 3. Performance Metrics



The figure illustrates the primary timing-based performance metrics obtained during system evaluation.

Figure 4. Metrics Summary

Metrics Summary	
Metric	Value
First Response Time	2.58 sec
Average Response Time	1.57 sec
Document Processing Time	175.88 sec
Total Queries Tested	15
Average Retrieval Score	0.99
Average Response Length	2077 chars

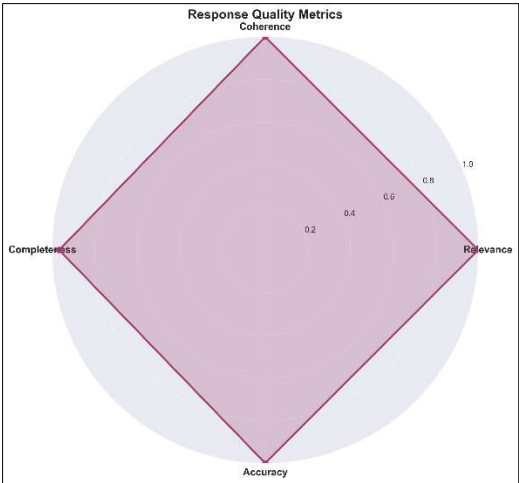
The figure presents a summary of the evaluation metrics obtained during system testing.

6.3 Retrieval and Response Quality

The retrieval and response evaluation results indicate that the proposed system was capable of generating context-aware educational responses supported by retrieved document content. The evaluation focused on retrieval relevance, response coherence, completeness, and overall response quality during educational question answering.

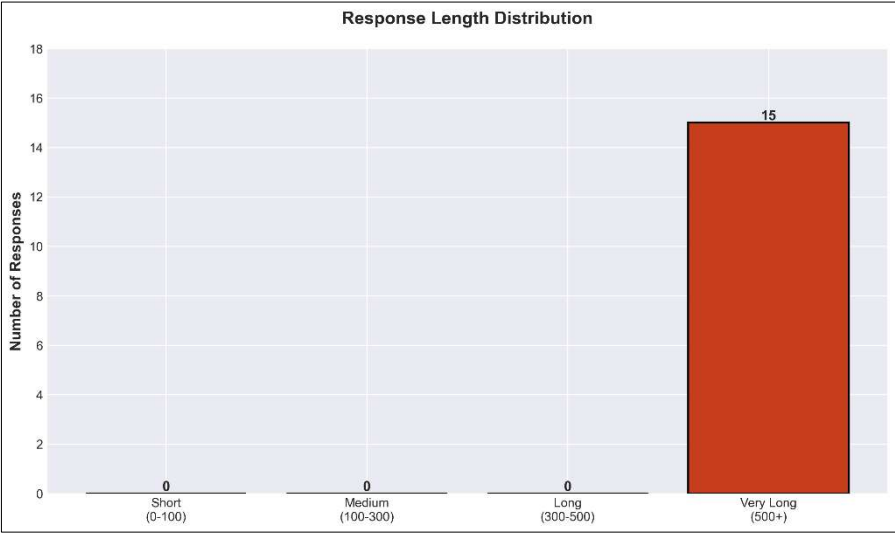
The system achieved an average retrieval score of 0.99 across the evaluated queries, indicating strong similarity-based retrieval performance during document search operations. In addition, the generated responses demonstrated consistent response quality and contextual relevance throughout the evaluation process.

Figure 5. Response Quality Metrics



The figure illustrates the evaluated response quality metrics during system testing.

Figure 6. Response Length Distribution

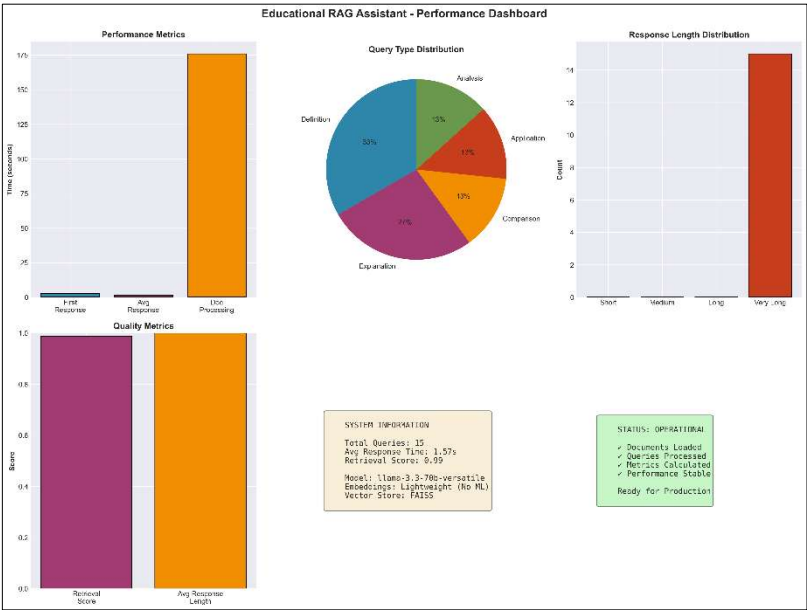


The figure presents the distribution of generated response lengths during system evaluation.

6.4 Visualization and Analysis

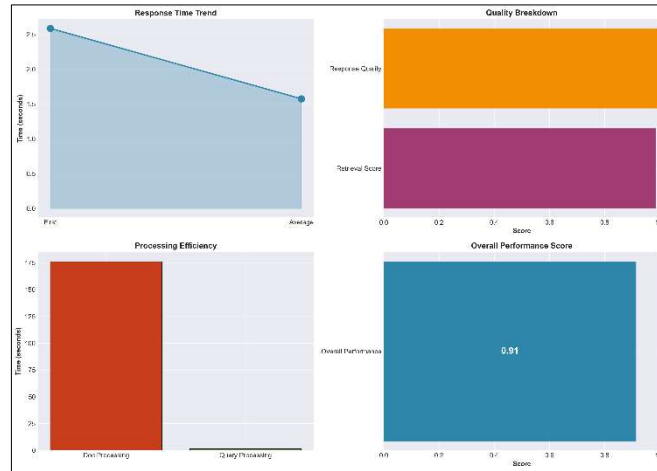
To support evaluation analysis and result interpretation, an integrated performance dashboard was developed to visualize system metrics, retrieval performance, query distribution, and response statistics within a unified interface.

Figure 7. Comprehensive Performance Dashboard



The dashboard presents an integrated overview of the system evaluation metrics and performance analysis.

Figure 8. Detailed Performance Analysis



The figure illustrates detailed performance analysis results related to timing, retrieval quality, and processing efficiency.

7. Prototype Interface and Deployment Description

7.1 Prototype Interface

The **EduRAG** system includes an interactive prototype interface developed using the Streamlit framework. The interface enables users to upload educational PDF documents, submit natural language queries, and receive generated responses through a unified educational question-answering environment.

The prototype interface was designed to simplify user interaction with the Retrieval-Augmented Generation pipeline by integrating document uploading, query submission, and response visualization within a single interface. In addition, the interface supports real-time interaction between users and the proposed educational assistant during document retrieval and response generation operations.

7.2 Deployment Description

The proposed **EduRAG** system was implemented and executed within a Python-based development environment. The deployment pipeline integrates document preprocessing, vector embedding generation, FAISS vector storage, retrieval operations, and large language model response generation within a unified Retrieval-Augmented Generation framework.

The system utilizes the Streamlit framework for interface execution, FAISS for vector-based similarity retrieval, and the Groq API service for large language model inference. During execution, uploaded educational documents are processed locally

before retrieval operations and response generation are performed through the integrated RAG pipeline.

Conclusion

This project presented **EduRAG**, a Retrieval-Augmented Generation system designed to support educational question answering using document-based response generation. The proposed system integrates document retrieval, vector-based similarity search, and large language model response generation within an interactive educational assistant.

The implemented system demonstrated effective retrieval performance and context-aware response generation during evaluation. In addition, the integration of Streamlit, FAISS vector storage, and the Groq language model service enabled the development of an interactive and practical educational prototype capable of handling multiple educational query types.

Overall, the proposed **EduRAG** system provides a practical approach for educational information retrieval and document-grounded question answering while demonstrating the effectiveness of Retrieval-Augmented Generation techniques in educational applications.

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